Mitochondrial State as a Gate on Conscious Neural Rendering

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Abstract

Neural computation is energetically expensive, but its physical support is neither spatially uniform nor temporally constant. Mitochondria differ across cells and subcellular compartments in ATP production, inner-membrane potential, calcium handling, redox state, position, and morphology. Primary experiments show activity-driven local ATP synthesis, calcium-sensitive metabolic tuning, activity-dependent mitochondrial repositioning, structural remodeling near synapses, complex-I- dependent changes in anesthetic sensitivity, and causal enhancement of long-term memory through mitochondrial calcium and metabolism. These results establish dynamic metabolic participation in neural function. The unresolved question is whether mitochondrial state selectively gates a named neural route, task epoch, and representational distinction rather than merely changing global capacity, health, or arousal. This paper formalizes that distinction. An ongoing brain-body state creates a baseline of mitochondrial readiness; phasic sensory, cognitive, or motor demand perturbs local compartments; the resulting ATP, membrane-potential, calcium, redox, position, and morphology state changes what a neural receiver integrates, inhibits, transmits, stabilizes, or learns. Twenty numbered equations define ATP balance, a six-variable readiness vector, receiver transformation, receiver-relative difference, content specificity, conditional information, mediation, matched rescue, dynamic state estimation, lagged influence, route selectivity, local-versus-global rescue, and a conjunctive causal gate. The operation is joined to Self Aware Networks, Neural Array Projection Oscillation Tomography, and Phase Wave Differentials only after the cell biology is established. The core model is downstream of a separate magnetic-induction hypothesis and remains intact if that actuator fails. Experiments combine compartment-resolved metabolic imaging, electrophysiology, phase and propagation analysis, cell-type-restricted perturbation, stage-specific behavior, and alternative- path rescue. Support requires local temporal precedence and route-specific rescue beyond matched global physiology. The result would establish mitochondria as structured participants in conscious neural rendering, without requiring mitochondria to store semantic content independently.
Keywords: mitochondria; neural metabolism; ATP; calcium; synaptic plasticity; neural rendering; receiver-state dependence; Phase Wave Differential; NAPOT; consciousness; causal mediation

1. The problem: a neural signal encounters local physical readiness

Recognizing a cup and adjusting one's grip recruits visual, proprioceptive, mnemonic, and motor pathways. Yet the cells carrying the signal are never in exactly the same state. One active dendritic or axonal compartment may be near a mitochondrion with strong membrane potential and available respiratory capacity. Another may have recent calcium load, altered redox state, lower local ATP availability, or a mitochondrion positioned farther from the active synapse. Similar input can therefore meet different capacities for integration, inhibition, vesicle cycling, ion- gradient restoration, and plasticity.

The familiar statement "neurons need energy" is too coarse to explain selection. A global ATP reduction impairs many functions. A selective gate is more demanding: after health, arousal, movement, temperature, oxygenation, and firing are controlled, a local mitochondrial variable must predict or causally change a specific receiver, route, epoch, content distinction, action, or report.

2. Established mitochondrial operations in neurons

2.1 Activity drives local ATP synthesis

Action potentials, ion pumping, transmitter cycling, and plasticity consume energy [1,2]. Using the Syn-ATP reporter, Rangaraju et al. showed that electrical activity imposes substantial presynaptic demand met through activity-driven glycolytic and mitochondrial ATP synthesis [3]. Brief interruptions in stimulated ATP production impaired presynaptic function. This is direct evidence for local dynamic supply-demand coupling. It does not identify represented content.

2.2 Calcium couples neural activity to mitochondrial response

Mitochondrial calcium changes metabolism, cytosolic calcium, release, and plasticity. Macaskill et al. identified Miro1 as a calcium sensor that links glutamatergic activity to mitochondrial localization [4]. Vaccaro et al. found that terminals containing mitochondria had reduced presynaptic calcium and transmitter release and that Miro1-dependent repositioning homeostatically rescaled calcium after prolonged activity changes [5]. Thus mitochondrial presence can damp or support a local signal depending on route and context; "more mitochondria" is not a universally positive direction.

Ashrafi et al. showed that brain-enriched MICU3 tunes axonal mitochondrial calcium sensitivity [6]. Removing MICU3 prevented activity-driven acceleration of local ATP synthesis and impaired presynaptic function under oxidative fuel conditions. This is a measured input -> organelle state -> synaptic-output chain.

2.3 Membrane potential and redox variables are coupled but distinct

Mitochondrial inner-membrane potential, $\Delta\Psi\_m$, supports ATP synthesis and transport while interacting with calcium, substrates, reactive oxygen species, and signaling. ATP concentration, ATP production, oxygen consumption, NADH/FAD state, ROS, calcium, and membrane potential are different observables [7]. The paper therefore preserves them separately rather than converting all metabolism into one hidden "energy" number.

2.4 Position and morphology change local capacity

Mitochondria move, stop, fuse, divide, change cristae, and alter organelle contacts. Rangaraju, Lauterbach, and Schuman showed that spatially stable dendritic mitochondrial compartments fuel local translation during synaptic plasticity; local depletion impaired stimulated spines without producing the same effect at remote spines [28]. Bapat et al. then identified VAP-dependent actin tethering that stabilizes mitochondria near postsynaptic spines and supports a bounded dendritic domain of plasticity [29]. These are direct precedents for spatially local support, not evidence that mitochondria encode semantic content. During homeostatic plasticity, Shah et al. report increased dendritic mitochondrial length, local area near spines, cristae surface and curvature, ER contacts, ribosomal recruitment, ATP-synthase clustering, mitochondrial ATP production, and spine ATP [8]. This 2025 source remains a preprint at the present audit date. It nevertheless supplies a measured structural-to-local-ATP hypothesis that final release must status-check again.

2.5 Mitochondrial state can alter memory and anesthesia-sensitive transmission

Zimin et al. found that isoflurane inhibits complex-I-dependent respiration and disrupts excitatory synaptic function under increased demand, with ATP, vesicle recycling, and adenosine signaling all implicated [9]. Ramadasan-Nair et al. identified circuit-specific NDUFS4 contributions to anesthetic sensitivity [10]. These results establish one mitochondrial route within a multimechanistic state transition.

Vishwanath et al. reduced the mitochondrial calcium exporter Letm1 in neurons [11]. Slower calcium efflux increased activity-linked presynaptic ATP, and PDP1 knockdown abolished the metabolic increase. Circuit-restricted manipulations enabled long-term memory in fly and mouse conditions where controls failed. This establishes a causal calcium -> metabolism -> long-term-memory route, not mitochondrial storage of memory content.

3. Three nested propositions

Proposition A: permissive capacity

Mitochondrial state constrains whether a compartment can sustain ordinary signaling. Severe impairment lowers function broadly. This well-supported proposition has low content specificity.

Proposition B: route- and epoch-specific gating

Local ATP, membrane potential, calcium, redox, position, or morphology changes the transformation of a named receiver in a defined task epoch. The effect remains after global firing, arousal, movement, temperature, oxygenation, and viability are controlled.

Proposition C: contribution to a rendered conscious distinction

The local gate changes a distributed pattern that predicts a specific perceptual, mnemonic, choice, action, or report distinction. Temporal precedence, mediation, and rescue establish the mitochondrial change as a causal member of the pathway rather than a consequence of neural activation. Evidence for A is not evidence for C; the paper is designed to decide B and C.

4. Development of the present proposal

Stage	Source-bound contribution	Date boundary
August 2011	self as a prediction pattern in the neocortex, whole-neocortex observer, and internally constructed world-model ancestry	owner-metadata-supported recording; current public page exposes a 2025 date; no mitochondrial atom assigned [30]
August 2012	recursive EEG-to-light/sound feedback, coordinated neural-wave language, embodied state change, and mind-as-projector architecture	live platform-dated public sources; no mitochondrial atom assigned [31,32]
2017	feedback-connected and connectionless communication analogy; deliberate Alternating Transmission Protocol/adenosine-triphosphate bridge	public Neural Lace Podcast, 11 April 2017 [13]
April 2021	ATP supply may constrain signaling/plasticity; stimulation can increase demand without restoring support	service-dated private multi-speaker recording; quotation requires audio readback [14]
November 2021	electrical, magnetic, mechanical, chemical, and metabolic carriers in a multiscale cellular system	service-dated private recording [15]
February 2022	metabolism and ionic/bioelectric computation as integrated embodied support	service-dated recording and later public Git family [16]
June 2022	firing/demand, proposed mitochondrial ATP response, receptor/protein plasticity, tonic/phasic activity, learning, flavin route, and consciousness joined publicly	immutable Git [17-20]
September 2023	metabolism-neural support loop, flavin measurement, and explicit quantum evidence boundaries	service-dated recording [21]
2024	molecular-to-oscillatory, plasticity, PWD, rendering, and consciousness bridge	published book [22]
February 2025	molecular-to-network agency, protein-state adaptation, ATP reinforcement, PWD, and rendered-pattern continuity are restated publicly; no six-variable mitochondrial gate is assigned	immutable public Git, 21 February 2025 [33]
2026	induction no-go correction, six-variable gate, content-specificity and rescue tests	current local synthesis [23]

The public June 2022 record goes beyond the statement that mitochondria supply energy. It joins local neural activity to a proposed metabolic response, receptor/protein plasticity, tonic/phasic context, learning, and conscious processing. The 2011 source remains authorship genealogy rather than strict public-priority proof because its present watch page does not expose the owner-recorded 2011 date. The two 2012 sources have live platform timestamps and establish recursive feedback, embodied state change, and projector architecture, but not a mitochondrial gate. The February 2025 Git source confirms continued ATP/protein/PWD architecture while leaving the six-variable, route-selective mitochondrial gate to the 2026 synthesis. The historical electromagnetic actuator is optional; the downstream metabolic-neural architecture does not depend on it.

5. The operation, before its vocabulary

An ongoing neural population maintains ion gradients, fuel delivery, temperature, oxygenation, and molecular repair. A new sensory, memory, or motor event raises demand in particular cells and compartments. Mitochondria respond according to current potential, calcium, redox state, substrate, position, and structure. That response changes integration time, release, ion-gradient restoration, inhibitory competition, and plasticity. The changed receiver alters what reaches the next population. Action and interoceptive consequences return as new evidence and update the next cycle.

Self Aware Networks (SAN) names this distributed receive-transform-project-feedback architecture. Neural Array Projection Oscillation Tomography (NAPOT) names the recurrent reconstruction and re- expression of partial neural relations across arrays. A Phase Wave Differential (PWD) is a receiver- relative difference between an event and the relation expected in that receiver's current context. Mitochondrial readiness is one proposed component of receiver state that can amplify, suppress, delay, stabilize, or write the effect into plastic structure.

Mitochondria need not contain a picture of the cup. Their state can change whether the cup-related route maintains, transforms, and transmits its distinction.

6. Independence from the magnetic-actuator branch

The separate manuscript Can an Action Potential Magnetically Drive Mitochondrial ATP Synthesis? tests one upstream actuator [23]. Its geometry-aware calculation makes ordinary Faraday induction negligible in the declared biological range. A flavin radical-pair mediator remains conditional and unvalidated. Activity-to-metabolism remains well supported through demand, calcium, transport, and substrate routes.

This paper begins from that intact branch:

neural and bodily state -> mitochondrial readiness -> neural transformation.

Spin-sensitive activation can be tested as one optional input. A failed magnetic experiment narrows the actuator and leaves ordinary mitochondrial gating intact.

7. Mathematical framework

7.1 Local ATP balance

For compartment i,

$$\begin{aligned}\frac{dA_i}{dt} &= P_i^{ox}(t)+P_i^{gly}(t)\\ &+J_i^{in}(t)-J_i^{out}(t)\\ &-C_i^{pump}(t)-C_i^{ves}(t)\\ &-C_i^{plast}(t)-C_i^{base}(t).\end{aligned} \tag{1}$$

This is a local bookkeeping interface for oxidative and glycolytic production, ATP or substrate exchange across the compartment boundary, and pump, vesicle, plasticity, and maintenance costs. The flux terms may represent transport, diffusion, or experimentally defined exchange with adjacent cytosol or compartments. No single ATP sensor identifies every term, so production, consumption, and exchange require independent constraints.

7.2 Six-dimensional readiness

Define

$$M_i(t)=[A_i(t),\Delta\Psi_{m,i}(t),C^m_i(t),R^{redox}_i(t),\\ P^{pos}_i(t),S^{morph}_i(t)]. \tag{2}$$

The components are ATP availability, inner-membrane potential, mitochondrial calcium, a declared redox vector, position relative to the active compartment, and morphology/network state.

Standardized variables are

$$\tilde{M}_{ik}(t)=\frac{M_{ik}(t)-\mu^{base}_{ik}}{\sigma^{base}_{ik}}. \tag{3}$$

A bounded task-specific capacity summary is

$$G_i(t)=\mathrm{logit}^{-1}\!\left(\beta_0+\Sigma^{6}_{k=1}\beta_k\tilde{M}_{ik}(t)+\Sigma_{k<\ell}\beta_{k\ell}\tilde{M}_{ik}(t)\tilde{M}_{i\ell}(t)\right). \tag{4}$$

G_i predicts capacity for a declared operation; it is not a consciousness score.

7.3 Receiver transformation

For input X_(ji), conventional receiver state R_i, mitochondrial state M_i, and context C_i,

$$Y_i(t+\delta)=F_i\!\left(X_{ji}(t),R_i(t),M_i(t),C_i(t)\right)+\varepsilon_i(t). \tag{5}$$

Adding M_i must improve held-out prediction and controlled intervention for a named output.

7.4 Expected relation and PWD

$$\mu_{ji}(t)=E\!\left[Y_i(t+\delta)\mid X_{ji}(t),R_i(t),C_i(t),H_t\right], \tag{6}$$

$$D_{ji}(t)=Y_i(t+\delta)-\mu_{ji}(t), \tag{7}$$

and

$$\Pi_{ji}(t)=\sqrt{D_{ji}(t)^{T}\Sigma_{ji}(t)^{-1}D_{ji}(t)}. \tag{8}$$

The signed vector D_(ji) preserves whether timing, phase, amplitude, duration, release, or propagation changed. A large Π_(ji) can represent novelty, adaptation, impairment, or useful differentiation; task context decides the interpretation.

7.5 Incremental information and content specificity

Mitochondrial state has incremental neural value when

$$I(Y;M| X,R,C)>0. \tag{9}$$

For task label K, a stronger content test is

$$I(K;M| X,R,C,A_{global})>0, \tag{10}$$

where global arousal, motion, temperature, oxygenation, viability, and firing are included in A_(global). Define

$$CSI=\Delta L_{matched\ content}-\Delta L_{matched\ arousal}. \tag{11}$$

A positive cross-validated specificity index supports route- or content-related value beyond an arousal-matched control. It does not require semantic content to be stored inside mitochondria.

7.6 Causal mediation and rescue

For restricted intervention U, mitochondrial mediator M, neural outcome N, and behavior/report B,

$$U\rightarrow M\rightarrow N\rightarrow B. \tag{12}$$

A task-epoch difference-in-differences contrast is

$$\Delta_{DiD}=(Y^{target}_{post}-Y^{target}_{pre})-(Y^{control}_{post}-Y^{control}_{pre}). \tag{13}$$

Matched rescue is

$$R_{rescue}=(Y^{rescue}-Y^{perturbed})-(Y^{sham}-Y^{perturbed}). \tag{14}$$

The complete path is summarized by

$$\Gamma_{M\rightarrow B}=\gamma_{U\rightarrow M}\gamma_{M\rightarrow N}\gamma_{N\rightarrow B},\quad \Pr(|\Gamma_{M\rightarrow B}|>\gamma_{\min})\geq 1-\alpha. \tag{15}$$

Each link remains separately reported so global impairment cannot masquerade as selection.

7.7 Dynamic identifiability, lag, route specificity, and causal gating

A static readiness score can correlate with neural activity even when mitochondria merely respond to that activity. The six variables are therefore treated as a measured dynamical state. For sampling interval Δ t,

$$M_i(t+\Delta t) = A_i M_i(t)+B_i^u u_i(t)+B_i^d d_i(t)+w_i(t),$$
$$O_i(t)=H_i M_i(t)+v_i(t), \tag{16}$$

where u_i is the restricted intervention, d_i contains measured neural demand, substrate, oxygenation, temperature, movement, and arousal, O_i is the vector of calibrated observations, H_i is the observation matrix, and w_i,v_i are declared process and measurement noise. The capacity summary G_i from Equation (4) and receiver context C_i from Equation (5) retain their earlier meanings; neither symbol is reused here. State dimensionality, priors, sampling rate, and observability checks are frozen before confirmatory analysis. Equation (16) prevents a single fluorescent proxy from being treated as the entire mitochondrial state.

For a preregistered lag window 0≤τ≤ L, the incremental mitochondrial contribution to neural outcome N_i is estimated as

$$N_i(t) = \alpha_i + \int_0^L K_{MN,i}(\tau)^T M_i(t-\tau),d\tau$$
$$+ \beta_i^T C_i(t) + \varepsilon_i(t). \tag{17}$$

The kernel K_(MN,i) must predict held-out neural data after conventional neural and body covariates C_i are included. Forward lag, reverse lag, shuffled time, demand matching, and a restricted intervention are jointly required. Equation (17) alone is predictive rather than causal; it cannot distinguish mitochondrial mediation from a shared response to demand.

A local gate should discriminate a target route or task epoch from a matched control. Let U=1 denote the mitochondrial intervention and U=0 sham. The route interaction is

$$\Delta_{route} = (E[Y_{target}| U=1]$$
$$- E[Y_{target}| U=0])$$
$$- (E[Y_{control}| U=1]$$
$$- E[Y_{control}| U=0]). \tag{18}$$

Target and control routes are matched for baseline firing, demand, anatomical scale, task difficulty, and measurement quality. The sign and minimum meaningful value are preregistered. A nonzero raw intervention effect with a null Δ_(route) supports global capacity, not route-selective gating.

The strongest rescue test contrasts restoration at the target compartment with a manipulation that improves global capacity while leaving that compartment unresolved. On one common outcome scale and relative to the same perturbation baseline, define

$$S_{local} = [(Y_T^{local} - Y_T^{pert}) - (Y_C^{local} - Y_C^{pert})] - [(Y_T^{global} - Y_T^{pert}) - (Y_C^{global} - Y_C^{pert})]. \tag{19}$$

A selective gate predicts a preregistered positive S_(local), equivalent restoration of the named mitochondrial mediator, and recovery of the target neural relation without a broad increase in firing, arousal, movement, temperature, or viability. Local and global rescues must be matched for overall ATP, viability, and measurement exposure. If global rescue performs equally well, the result is classified as permissive capacity.

No aggregate score may conceal a missing causal link. For binary gates frozen before confirmatory analysis,

$$G_{mito} = g_{measurement}, g_{order}, g_{U\rightarrow M}, g_{M\rightarrow N}, g_{route}, g_{blockade}, g_{local}, g_{N\rightarrow B}, g_{replication}. \tag{20}$$

A complete route-specific mitochondrial-to-rendering result requires G_(mito)=1. This is a conjunctive decision rule, not a posterior probability that the whole theory is true. If one gate fails, the valid lower-level result is retained and the unsupported link is named.

8. Variable, direction, timescale, and compartment

Variable	Measured precedent	Possible direction	Candidate role	Required measurement
ATP availability/production	activity-driven local synthesis [3]; Letm1/PDP1 memory chain [11]	insufficient, matched, or accumulated	pumps, vesicle cycling, repeated plasticity	compartment ATP plus production/consumption estimate
inner-membrane potential	coupled to ATP, transport, calcium, and stress [7]	excessive or collapsed values can both be maladaptive	rapid capacity and slower stress signal	calibrated reporter with photophysical controls
mitochondrial calcium	MICU3 and Letm1 studies [6,11]	uptake/retention can raise metabolism; overload can harm	activity-metabolism transducer	mitochondrial and cytosolic calcium together
redox state	mitochondrial signal-transduction literature [7]	physiological signal or oxidative damage	protein/channel state and transcription	specified NADH/FAD/ROS reporters
position	Miro1 studies [4,5]; spatially bounded translation/plasticity support [28,29]	local buffering can lower calcium/release while supporting homeostasis	route-specific gain, buffering, and local support	live organelle tracking at identified synapses
morphology	Shah et al. 2025 [8]	remodeling may expand sustained local capacity	plasticity-scale ATP domain	fusion/fission, cristae, contacts, volume, ATP

Every content claim must declare variable, direction, compartment, and timescale.

9. Experimental program

9.1 Compartment-resolved observation

Measure the six variables with voltage, spikes, release, phase, propagation, and behavior in a task with repeated stimuli and a predefined content distinction. Train Equations (5)-(11) on one set and test held-out sessions.

9.2 Restricted perturbation

Manipulate one mitochondrial operation in a defined cell type and compartment during encoding, delay, choice, motor, or report. Confirm that the mitochondrial mediator changes before the neural outcome. Acute and chronic interventions answer different questions and remain separate.

9.3 Two matched rescues

Restore the mediator through a different route while holding general health as constant as possible. Examples include restoring local ATP capacity without restoring the original calcium manipulation, or restoring position without globally increasing metabolism. A second negative control deliberately improves global ATP while leaving the target compartment or route unchanged. Broad recovery supports capacity; selective route rescue supports gating.

9.4 Coherence, splay, propagation, and PWD

Test whether readiness changes structured phase alignment, stable phase separation, inhibitory routing, receiver-relative timing, or traveling-wave propagation after firing and arousal are matched. More coherence is not automatically better. Coupling and dispersion can each preserve useful distinctions depending on route and task.

9.5 Memory stages

Separate encoding, consolidation, retrieval, reconsolidation, and report. Vishwanath et al. supply a causal metabolic-memory precedent [11]; content and conscious-rendering claims require neural and behavioral stage separation beyond that result.

9.6 Anesthesia and state transition

Track all variables across wakefulness, sedation, loss and recovery of responsiveness, and sleep. Compare agents with different molecular actions and include motor-suppression controls. Complex I is one contribution [9,10], not a universal consciousness switch.

9.7 Action and interoceptive return

Follow sensory evidence through candidate action, motor output, and returned visual, proprioceptive, and interoceptive evidence. Test whether the gate changes the selected route before action and whether the consequence updates the next neural and mitochondrial state.

9.8 Confirmatory preregistration matrix

Each experiment freezes its measurement and decision structure before confirmatory acquisition:

Stage	Experimental unit	Primary variables	Confirmatory contrast	Capacity control	Selective-gate pass condition
state estimation	identified soma, dendrite, axon, terminal, or circuit	ATP, $\Delta\Psi$_m, mitochondrial calcium, declared redox measures, position, morphology	dynamic model versus covariate-only model	reporter calibration, temperature, oxygen, substrate, motion	observable state with held-out predictive value
temporal direction	repeated task event in the same unit	M_i, voltage/spikes/release, phase/propagation	preregistered forward lag	reverse lag, time shuffle, and demand-matched null	mitochondrial mediator precedes the named neural consequence
route perturbation	cell-type- and compartment-restricted intervention	mediator, target route, matched route	Equation (18)	global physiology and demand matched	predicted route interaction without sickness or broad suppression
rescue	perturbed preparation	target-compartment and global-capacity rescues	Equation (19) on a common scale	equal overall ATP, viability, and exposure	local rescue exceeds global rescue for the target relation

Stage	Experimental unit	Primary variables	Confirmatory contrast	Capacity control	Selective-gate pass condition
content/action	preregistered epoch and task	neural distinction, choice/action/report	mediated target versus matched-content contrast	arousal, difficulty, movement, report, sensory and motor controls	specific M→ N→ B chain
replication	independent batch, cohort, or laboratory	frozen primary outcomes	same model and thresholds	complete null and adverse-event reporting	directionally consistent corrected estimate

The protocol freezes randomization, blinding, unit of analysis, sample-size rationale, sensor cross-talk tests, missing-data handling, lag window, state-space dimension, estimator, equivalence bounds, multiple-testing correction, exclusions, and stopping rule. Candidate discovery remains exploratory until repeated with the frozen design.

10. Longitudinal comparison with the 2026 circuit-readiness synthesis

Sandi et al. describe two complementary modes of mitochondrial support: a baseline mode sustaining circuit architecture and physiological properties over long timescales, and an activity-evoked mode in which local mitochondrial outputs support transmission and plasticity [12]. They explicitly frame circuits as differing in readiness for recruitment and link mitochondrial state to engagement, adaptation, cognition, and behavior.

The operational and narrative architecture closely approaches this paper's tonic readiness -> phasic demand -> local mitochondrial response -> changed receiver and route sequence. Micah's immutable June 2022 sources are earlier on the narrower joined metabolic-response, receptor/protein- plasticity, tonic/phasic, learning, and consciousness architecture. The external field is much older on energy, calcium, transport, plasticity, stress, and behavior, and Sandi's own pre-2022 work on stress/anxiety and mitochondria must be preserved in the atom-level genealogy.

The current source-bound disposition is therefore:

- external researchers earlier on many mitochondrial constituent atoms;
- Micah publicly earlier in June 2022 on his joined tonic/phasic metabolic-to-plasticity-to- consciousness operator;
- Sandi et al. later in July 2026 on the explicit baseline/activity-evoked circuit-readiness synthesis;
- content-specific receiver/PWD/rendering equivalence remains a partial-match question pending full body and citation-genealogy audit;
- chronology and similarity establish no access or intent conclusion.

Picard and Shirihai's broad mitochondrial signal-transduction review was published in November 2022 [7], after Micah's June Git stage but as a synthesis of older science. Publication timing does not assign its older constituent discoveries to 2022. The paper's bounded residual is the joined, receiver-relative and content-tested SAN operator.

11. Alternative explanations and controls

Principal alternatives are global energetic impairment, nonspecific firing reduction, temperature, oxygenation, vascular change, pH, movement, stress, arousal, phototoxicity, reporter cross-talk, extra-mitochondrial calcium/redox effects, and developmental compensation. Designs therefore pair global and compartment ATP; mitochondrial and neuronal membrane potential; mitochondrial and cytosolic calcium; chemically specific redox reporters; viability and ion-gradient integrity; temperature, oxygen, substrate, blood flow, and movement; arousal- and difficulty-matched tasks; acute/chronic timing; sham intervention; alternative-path rescue; and preregistered variables with multiple-testing correction.

12. Failure conditions

The selective-gating hypothesis is narrowed or rejected when:

1. mitochondrial variables add no held-out value beyond conventional neural/body covariates;
2. the mitochondrial change follows rather than precedes the neural event;
3. perturbation produces only sickness, sedation, arousal change, or lower firing;
4. route, compartment, epoch, and content do not survive matched controls;
5. the named mitochondrial mediator does not explain the neural effect;
6. matched rescue fails while a nonmitochondrial pathway rescues the outcome;
7. neural and behavioral effects persist contrary to mediator blockade/restoration;
8. PWD, phase, splay, or propagation adds no value beyond established models;
9. calcium, redox, or vascular change outside the proposed route fully explains the result.

Failure of the magnetic actuator is not failure of this downstream model. Failure of content specificity demotes the result from selective rendering to permissive metabolic capacity.

13. Conclusion

Mitochondria already supply local ATP, handle calcium, reposition at active synapses, change redox signaling, remodel with demand, influence plasticity and memory, and contribute to anesthetic sensitivity. The remaining scientific step is to decide whether these operations selectively gate a particular neural transformation.

The framework preserves six variables, requires out-of-sample value beyond conventional neural covariates, distinguishes content specificity from global activation, and demands temporal precedence, mediation, and matched rescue. In SAN terms, mitochondrial state becomes part of neural computation when it changes what a receiver does with an input and thereby changes a PWD, a routed population pattern, and a returned perceptual or action consequence. The conscious system is the recurrent brain-body network whose cells, organelles, signals, actions, and feedback continually change one another.

Author contributions and AI-assisted development

Micah Blumberg conceived the theory, defined the mechanism and research questions, supplied the historical source corpus, and is the manuscript author. AI tools assisted with source routing, mathematical formalization, drafting, and internal consistency checks. Blumberg reviewed the scientific claims and retains responsibility for the manuscript. Historical sources, external evidence, current synthesis, and proposed experiments are identified separately.

Data and materials availability

This theoretical and methods paper reports no new participant, animal, or wet-laboratory data. Its governed source manifest, claim-source genealogy atlas, validation receipts, and figure specifications accompany the manuscript package. External sources are identified by DOI or stable public URL. Private development sources are used only for authorship genealogy and are not presented as public priority evidence.

Ethics statement

The manuscript proposes future cellular, neural, animal, and behavioral experiments. Any study involving animals or human participants will require prospective institutional review, welfare protections, consent where applicable, and a preregistered protocol. No such experiment is reported here.

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